Task 1 If T is linear? bases for $N(T)$ and $R(T)$, nullity and rank of T verify dimension theorem, determine T is one-to-one or onto.

$T: \mathbb{R}^3 \to \mathbb{R}^2$, $T((a_1, a_2, a_3)) = (a_1 - a_2, 2a_3)$

$T: \mathbb{R}^6 \to \mathbb{R}^4$, $T(a_1, a_2, a_3, a_4, a_5, a_6) = (2a_1 - a_2, a_3 + a_4, 0, 0)$

$T: M_{1\times 2}(\mathbb{R}) \to M_{1\times 3}(\mathbb{R})$, $T([a_1, a_2]) = T([a_1 + a_2, 0, 2a_1 - a_2])$

① $T((v_1, v_2, v_3) + (u_1, u_2, u_3))$

$= T((v_1+u_1, v_2+u_2, v_3+u_3)) = ((v_1+u_1-v_2-u_2), (2v_3+2u_3))$

$= (v_1 - v_2, 2v_3) + (u_1 - u_2, 2u_3) = T((v_1, v_2, v_3)) + T((u_1, u_2, u_3))$

$T(c(v_1, v_2, v_3)) = T((cv_1, cv_2, cv_3)) = (cv_1 - cv_2, 2cv_3)$
$= c(v_1 - v_2, 2v_3) = cT((v_1, v_2, v_3))$

let $T(S) = (0, 0)$, $S = (s_1, s_2, s_3)$, $S \in \mathbb{R}^3$

$s_1 - s_2 = 0$, $2s_3 = 0$

$s_1 = s_2$, $s_3 = 0$

☆ $N(T) = \text{span}\{(1, 1, 0)\}$, nullity of $T = 1$

☆ $R(T) = \mathbb{R}^2$, rank of $T = 2$

☆ nullity of T plus rank of $T = 1 + 2 = 3 = \dim(\mathbb{R}^3)$

☆ T is not one-to-one. T is onto

☆ T is linear

② let $A, B \in \mathbb{R}^6$

$A = (a_1, a_2, a_3, a_4, a_5, a_6)$

$B = (b_1, b_2, b_3, b_4, b_5, b_6)$

$T(A+B) = (a_1+b_1, a_2+b_2, a_3+b_3 \cdots\cdots) = (2a_1+2b_1-a_2-b_2, a_2+a_3+b_2+b_3, 0, 0)$

$= (2a_1-a_2, a_3+a_2, 0, 0) + (2b_1-b_2, b_3+b_2, 0, 0)$

$= T(A) + T(B)$

$T(cA) = T(c(a_1, a_2, a_3, \cdots)) = (2ca_1 + ca_2, ca_3 + ca_2, 0, 0)$

$= c(2a_1+a_2, a_3+a_2, 0, 0) = cT(A)$

☆ T is linear

let $T(S) = (0,0,0,0)$, $S \in \mathbb{R}^6$

$S = (s_1, s_2, s_3, s_4, s_5, s_6)$

$$\begin{cases} 2s_1 - s_2 = 0 \\ s_2 + s_3 = 0 \end{cases} \qquad s_1 : s_2 : s_3 = 1 : 2 : -2$$

☆ $N(T) = \text{span}\{(1,2,-2,0,0,0), (0,0,0,1,0,0), (0,0,0,0,1,0), (0,0,0,0,0,1)\}$

☆ nullity of $T = 4$

☆ $R(T) = \text{span}\{(1,0,0,0), (0,1,0,0)\}$

☆ rank of $T = 2$

☆ nullity of T plus Rank of T $= \dim(S) = 6$

☆ T is not one-to-one nor onto

(3) let $A, B \in M_{1\times 2}$ $A = [a_1, a_2]$, $B = [b_1, b_2]$

$T(A+B) = T([a_1+b_1, a_2+b_2]) = [a_1+a_2+b_1+b_2, 0, 2a_1+2b_1-a_2-b_2]$

$= [a_1+a_2, 0, 2a_1-a_2] + [b_1+b_2, 0, 2b_1-b_2]$

$= T(A) + T(B)$

let $c \in \mathbb{R}$

$T(cA) = T([ca_1, ca_2]) = [ca_1+ca_2, 0, 2ca_1-ca_2]$

$= c[a_1+a_2, 0, 2a_1-a_2]$

$= cT(A)$

* T is linear

let $S \in M_{1\times 2}$ $T(S) = [0, 0, 0]$

$S = [s_1, s_2]$

$\begin{cases} s_1+s_2=0 \\ 2s_1-s_2=0 \end{cases}$ $\quad s_1=0$, $s_2=0$

* $N(T) = (0,0)$, nullity of $T = 0$
* $R(T) = \text{span}\{(1,0,0),(0,0,1)\}$, rank of $T = 2$
* nullity of T plus rank of T = $\dim(S) = 2$

T is one-to-one but not onto

Task 2

① let $A, B \in \mathbb{R}^2$

$A = (a_1, a_2)$, $B = (b_1, b_2)$

$T(A+B) = T((a_1+b_1, a_2+b_2)) = (1, a_2+b_2)$

$T(A)+T(B) = (1, a_2) + (1, b_2) = (2, a_2+b_2)$

let $c \in \mathbb{R}$

$T(cA) = T((ca_1, ca_2)) = (1, ca_2)$

$cT(A) = c(1 . a_2) = (c, ca_2)$

$\because T(A+B) \neq T(A)+T(B)$, $cT(A) \neq T(cA)$

$\therefore T$ is not linear

③ let $A, B \in \mathbb{R}^2$ $A = (a_1, a_2)$, $B = (b_1, b_2)$

$T(A+B) = T((a_1+b_1, a_2+b_2)) = ((a_1+b_1), (a_1^2+2a_1b_1+b_1^2))$

$T(A)+T(B) = (a_1, a_1^2) + (b_1, b_1^2) = (a_1+b_1, a_1^2+b_1^2)$

let $c \in \mathbb{R}$

$T(cA) = T((ca_1, ca_2)) = (ca_1, c^2a_1^2)$

$cT(A) = c(a_1, a_1^2) = (ca_1, ca_1^2)$

$\because T(A+B) \neq T(A)+T(B)$, $cT(A) \neq T(cA)$

$\therefore T$ is not linear

③

~~$T(a_1$~~

let $A = (a_1, a_2),\ a_1 < 0,\ B = (b_1, b_2),\ b_1 > 0$

$T(A+B) = T((a_1+b_1, a_2+b_2)) = (|a_1+b_1|, a_2+b_2)$

$T(A) + T(B) = (-a_1, a_2) + (b_1, b_2)$

$= (b_1 - a_1, a_2+b_2)$

$T(A+B) \neq T(A) + T(B)$

T is not linear

Task 3

$T(2,3) = 3 \cdot T(1,1) - T(1,0)$

$= 3 \cdot (2,5) - (1,4)$

$= (6,15) - (1,4)$

$= (5,11)$

Base on dimension theorem

$\text{nullity}(T) + \text{rank}(T) = 2$

$R(T) = \text{span}\{T(1,0), T(1,1)\}$, $T(1,0)$ and $T(1,1)$ are linear independent

$\text{rank}(T) = 2$

$\text{nullity}(T) = (0,0)$

$\therefore$ T is one-to-one

Task4

If T is linear

$cT(A) = T(cA)$, $A \in \mathbb{R}^3$, $c \in \mathbb{R}$

$T(3,6,3) = 3T(1,2,1)$

$(2,1) \neq 3(1,1)$

T is not linear

~~Task5~~

$\cancel{N(T) = \{(x,y) \in \mathbb{R}^2 \text{ where } x=0\}}$

$\cancel{T(0,a) = (0,0),\ a \in \mathbb{R}}$

$\cancel{T(0,b) = (0,0),\ b \in \mathbb{R}}$

~~let~~ $\cancel{T(x,y) = (x,xy)}$

$\cancel{T(a_1+b_1,\ a_2+b_2) =}$

Task 5

$T : R^2 \to R^2$

$N(T) = \{(x, y) \in R^2 \text{ where } x = 0\}$

$= \text{span}\{(0, 1)\}$

nullity of $T = 1$

base on dimensiot theorem, rank $= 1$

~~$R(T) = \text{span}\{(a_1, a_2)\}$, $a_1, a_2 \in R$~~

~~let $\beta = (1, 0)$, $T(\beta) = (1, 0)$~~

$R(T) = \text{span}\{T(1, 1)\}$

let $T(1, 1) = (1, 0)$, $T(x, y) = (x, 0)$

Test if T is linear:

$T(a_1 + b_1, a_2 + b_2) = (a_1 + b_1, 0) = (a_1, 0) + (b_1, 0)$

$= T(a_1, a_2) + T(b_1, b_2)$

$T(ca_1 + ca_2) = (ca_1, 0) = c(a_1, 0) = cT(a_1, a_2)$

T is linear

☆ $T(x, y) = (x, 0)$